Chromatic Numbers of the Plane with Two Forbidden Distances: a Seven-Color Lower Bound and Smaller Six-Chromatic Graphs

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October 2026

Abstract

Let $\chi\big(\mathbb{R}^2; D\big)$ denote the minimum number of colors required to color the plane so that no two points whose distance lies in D receive the same color, and set $d = (\sqrt{6}+\sqrt{2})/2 = 2\cos(\pi/12)$, so that $d^2 = 2+\sqrt{3}$ and $d^2 - 1 = \sqrt{2}\,d = 1+\sqrt{3}$. Our main result is

$$\chi\big(\mathbb{R}^2; \{1, d\}\big) \;\geq\; 7$$

for arbitrary colorings, with no measurability assumed (the averaging uses an invariant mean on the amenable group $E(2)$, a theorem of ZFC): for every proper 6-coloring, invariant monochromatic-pair densities $p_{\sqrt{2}}$ and $p_{1+\sqrt{3}}$ would have to satisfy $p_{\sqrt{2}} + p_{1+\sqrt{3}} \geq 1$ (an 8-point pigeonhole gadget of Parts) and $p_{\sqrt{2}}, p_{1+\sqrt{3}} \leq 5/11$ (two explicit 12-point chain configurations whose long-range pairs land exactly on the forbidden distances via the identity $d^2 - 1 = \sqrt{2}\,d$), a contradiction since $1 \leq 10/11$ is false. The argument realizes a 2019 sketch of J. Parts in a Polymath16 blog comment; the contribution here is a fully explicit finite form with an explicit gap ($p_r \leq 5/11 = \frac{1}{2} - \frac{1}{22}$), each of the three finite statements carrying a DRAT proof trace checked by `drat-trim`, and the full 32-point geometry certified in exact symbolic arithmetic over explicit quadratic towers. The argument does not use the measurable-to-arbitrary transfer of OpenAI's 2026 proof that $\chi(\mathbb{R}^2) \geq 6$.

We also give, with exact-arithmetic edge sets and DRAT-certified non-colorability: a 26-vertex 6-chromatic $\{1, d\}$-graph (to our knowledge the smallest 6-chromatic two-distance graph known for any ratio, improving 31 [Parts 2020] and, for this ratio, 117 [De Neve et al. 2025]); a 30-vertex 6-chromatic $\{1, \sqrt{3}\}$-graph (previously 33); the six-coloring continuum $X_6 \supseteq [0.354, 0.667504532081]$ (previously $[0.354, 0.657]$), by 59 exact rational tilings; and the edge bound $e_5 \leq 2257$ for 5-chromatic unit-distance graphs, via a 2257-edge non-4-colorable spanning subgraph of the 509-vertex record graph (previously 2406). In the other direction, an explicit periodic 5-coloring of the plane by 4230 convex pieces whose random unit segment is monochromatic with probability at most 0.0031083 (certified in ball arithmetic) shows that every 6-chromatic unit-distance graph has at least 322 edges and 62 vertices (previously 182 and 42). As by-products we record forced monochromatic densities $p_{\sqrt{3}} \geq 1/28$, $p_d \geq 1/37$ and $p_{1/\sqrt{3}} \geq 1/67$ for invariant unit-distance 5-colorings, together with elementary color-count-free density caps such as $p_{1/\sqrt{3}} \leq 1/3$; since $\chi(\mathbb{R}^2) \geq 6$ is now known, these statements about 5-colorings are vacuous, and we keep them as finite combinatorial statements. All certificates and verifiers are included in the accompanying `certificates/` folder (Appendix A).

1 Introduction

The Hadwiger–Nelson problem asks for the chromatic number of the plane, $\chi(\mathbb{R}^2) = \chi\big(\mathbb{R}^2; \{1\}\big)$: the least number of colors so that no two points at distance 1 share a color. After more than half a

century at $4 \le \chi(\mathbb{R}^2) \le 7$, de Grey [8] proved $\chi(\mathbb{R}^2) \ge 5$ with a 1581-vertex unit-distance graph, launching the Polymath16 project [23] and a rapid sequence of smaller witnesses [14, 11, 22].

In the *two-distance* variant one forbids two distances simultaneously: $\chi\big(\mathbb{R}^2;\{1,d\}\big)$ is the chromatic number of the graph on $\mathbb{R}^2$ with edges between points at distance 1 *or* d. Finite 6-chromatic witnesses were known for exactly four ratios: the golden ratio $(1+\sqrt{5})/2$ (Huddleston [18], without an explicit graph; an explicit 126-vertex graph was given by de Grey in Polymath16 [23], and the 31-vertex record is due to Parts [21]), $d = 2$ (Exoo–Ismailescu [12], reduced to 397 vertices by Parts [21]), and $d = \sqrt{3}$ and $d = (\sqrt{6}+\sqrt{2})/2$ (De Neve, Vanden Kerchove, Colle, Tavernier, and Pickavet [10], with 33 and 117 vertices respectively; the lower bound 6 for these two ratios had been obtained earlier by Pálvölgyi and Ágoston with the probabilistic method [19]). For no finite set D of distances was a bound $\chi\big(\mathbb{R}^2;D\big) \ge 7$ published, although J. Parts sketched a probabilistic argument for exactly this in a 2019 Polymath16 comment [20].

Note added (October 2026). The results of this paper were obtained in August 2026. Since then OpenAI [17] proved $\chi(\mathbb{R}^2) \ge 6$, so every two-distance chromatic number of the plane is at least 6, the statements about 5-colorings in Sections 6.3 and 7 are vacuous, and the win condition of Section 7 is moot. The explicit small 6-chromatic graphs of Section 5 are unaffected. The transfer principle of [17] extends to countable sets of algebraic distances (Crocker [7], conditionally on a theorem of [17], and the companion manuscript [4], with a Lean formalization), giving $\chi\big(\mathbb{R}^2;D\big) \ge 1/m_D(\mathbb{R}^2)$, where m_D is the maximal density of a measurable set avoiding D. The companion manuscript [3] combines this with Fourier density certificates and obtains, among other things, $\chi\big(\mathbb{R}^2;\{1,d'\}\big) \ge 7$ for every algebraic $d' \in [1.77, 1.92] \cup [7,\infty)$; with Bukh's theorem [6] one also gets $\chi\big(\mathbb{R}^2;\{1,N\}\big) \ge 17$ for all sufficiently large algebraic N, with no explicit threshold. The value $d = (\sqrt{6}+\sqrt{2})/2 \approx 1.932$ lies outside the ranges covered there, and Theorem 1.1 is proved differently: by three finite SAT-certified configurations and an invariant mean, without the transfer. Euclidean Ramsey consequences of the transfer are developed in [5].

All searches and certificates described below were computer-assisted. Throughout, $d := (\sqrt{6}+\sqrt{2})/2 = 2\cos(\pi/12) \approx 1.93185$, whose relevant algebra is

$$d^2 = 2+\sqrt{3}, \qquad d^2 - 1 \;=\; 1+\sqrt{3} \;=\; \sqrt{2}\,d. \tag{1}$$

Theorem 1.1 (Main theorem; invariant-coloring framework)**.** $\chi\big(\mathbb{R}^2;\{1,d\}\big) \ge 7$. *More precisely, there is no proper* 6*-coloring* $c\colon \mathbb{R}^2 \to \{1,\dots,6\}$ *avoiding monochromatic pairs at distances* 1 *and* d*: for any such* c*, invariant monochromatic-pair densities* $p_{\sqrt{2}}(c)$ *and* $p_{1+\sqrt{3}}(c)$ *exist (via an invariant mean on the isometry group) and satisfy the incompatible system*

$$p_{\sqrt{2}} + p_{1+\sqrt{3}} \;\ge\; 1, \qquad p_{\sqrt{2}} \;\le\; \tfrac{5}{11}, \qquad p_{1+\sqrt{3}} \;\le\; \tfrac{5}{11}.$$

The three displayed inequalities come from three finite configurations ($8 + 12 + 12 = 32$ points in total), each certified by a DRAT proof trace (`certificates/seven-colors/`; Section 3 and Appendix A). The framework, its provenance in ZFC, and its limitations are discussed in Sections 2–3; a counting blueprint that would replace the framework by a single finite graph is given in Section 4.

The remaining results rest on finite statements with exact-arithmetic and DRAT certificates:

Theorem 1.2.

(a) There are 26 *points in the plane spanning* 85 *pairs at distance exactly* 1 *or exactly* d *whose* $\{1,d\}$*-graph has chromatic number* 6 *and is vertex-critical. To our knowledge this is the smallest* 6*-chromatic two-distance graph known for any distance pair (previously* 31 *[21]; for this pair,* 117 *[10]).*

(b) There are 30 *points spanning* 100 *pairs at distance exactly* 1 *or exactly* $\sqrt{3}$ *whose graph has chromatic number* 6 *and is vertex-critical (previously* 33 *[10]).*

(c) The record 509*-vertex* 5*-chromatic unit-distance graph [22, 9] contains a spanning subgraph with* 2257 *edges that is not* 4*-colorable. Hence* $e_5 \le 2257$ *(previously* $e_5 \le 2406$ *[9]).*

(d) Every 6*-chromatic unit-distance graph has at least* 322 *edges and at least* 62 *vertices:* $e_6 \ge 322$ *and* $v_6 \ge 62$ *(previously* $e_6 \ge 182$, $v_6 \ge 42$ *[9]).*

(e) For every proper unit-distance 5*-coloring of the plane and every invariant mean, the monochromatic-pair densities of Definition 2.1 satisfy* $p_{\sqrt{3}} \ge 1/28$, $p_d \ge 1/37$ *and* $p_{1/\sqrt{3}} \ge 1/67$. *(Since* $\chi(\mathbb{R}^2) \ge 6$ *[17], no such coloring exists; the content is the underlying finite statement, see Section 6.3.)*

Each item is backed by files in `certificates/`: (a) by `g26/`, (b) by `g30/`, (c) by `e5-2257/`, (d) by `e6-lower-bound/`, and (e) by `g30/`, `g26/` and `forced-density/` respectively.

Quantity	Previous	Here	Certificate
Smallest 6-chromatic two-distance graph, any ratio	31 [21]	**26**	exact edges + DRAT (`g26/`)
… for $d = (\sqrt{6}+\sqrt{2})/2$	117 [10]	**26**	exact edges + DRAT (`g26/`)
… for $d = \sqrt{3}$	33 [10]	**30**	exact edges + DRAT (`g30/`)
$\chi\big(\mathbb{R}^2; \{1, (\sqrt{6}+\sqrt{2})/2\}\big)$	≥ 6 [19, 10]; ≥ 7 sketched [20]	$\ge$ **7**	exact geometry + 3 DRAT (`seven-colors/`)
Edges of the 509-vertex 5-chromatic UD graph	≤ 2406 [9]	$\le$ **2257**	DRAT (`e5-2257/`)
Edges / vertices of any 6-chromatic UD graph	≥ 182 / ≥ 42 [9]	$\ge$ **322** / $\ge$ **62**	exact tiling + ball arithmetic (`e6-lower-bound/`)
Six-coloring continuum X_6	$[0.354, 0.657]$ [15]	$[0.354, \mathbf{0.6675045}\ldots]$	59 exact rational tilings (`six-coloring-continuum/`)
Forced densities $p_{\sqrt{3}}$, p_d, $p_{1/\sqrt{3}}$ (vacuous since [17])	—	**1/28**, **1/37**, **1/67**	DRAT (`g30/`, `g26/`, `forced-density/`)

Attribution. The $\chi \ge 7$ argument is, in its skeleton, due to Jaan Parts, who sketched it on 24 December 2019 in comment 24686 on the fifteenth Polymath16 thread [20] and, to our knowledge, never published it. Our contribution is (i) fully explicit finite certificates — the specific 8- and 12-point configurations, their exact geometry, and their SAT/DRAT verification — (ii) the explicit constant 5/11 for the upper bound on p_r, and (iii) a counting blueprint toward a choice-free finite witness (Section 4). The record graphs of Theorem 1.2(a,b) were seeded from the published coordinates of De Neve et al. [10].

Organization. Section 2 sets up two-distance graphs and the invariant-coloring framework. Section 3 proves Theorem 1.1. Section 4 describes the counting blueprint toward a fully finite

witness, together with computational observations about it. Section 5 presents the record graphs with complete coordinate tables and verification recipes. Section 6 contains the unit-distance results (Theorem 1.2(c,d)). Section 7 develops finite caps valid for any number of colors, a bipartiteness obstruction at the distances $\sqrt{3}$ and 2, and the (now moot) win condition for $\chi(\mathbb{R}^2) \geq 6$. Section 8 extends the six-coloring continuum. Section 9 collects open problems. Appendix A documents the artifact bundle.

2 Preliminaries

2.1 Two-distance graphs

For a finite $P \subset \mathbb{R}^2$ and a set D of positive reals, let $G(P; D)$ be the graph on P with uv an edge iff $|u - v| \in D$. We write $\chi\big(\mathbb{R}^2; D\big)$ for the chromatic number of the (infinite) graph $G(\mathbb{R}^2; D)$ and abbreviate $G(P) = G(P; \{1, d\})$ when $D = \{1, d\}$ is understood. By the de Bruijn–Erdős theorem (with choice), $\chi\big(\mathbb{R}^2; D\big) \geq k$ holds iff some finite P has $\chi(G(P; D)) \geq k$; part of the interest of Theorem 1.1 is precisely that it does *not* yet come with such a finite witness (see Section 3.5).

All configurations in Section 5 have coordinates in a real quadratic-tower field ($\mathbb{Q}(\sqrt{3}, \sqrt{5})$ or $\mathbb{Q}(\sqrt{3}, \sqrt{7})$) with denominators dividing 8, so squared distances are computed in exact integer arithmetic and edge membership is decidable with no floating point. The $\chi \geq 7$ configurations of Section 3 are likewise exact algebraic configurations, certified end-to-end by symbolic computation in explicit quadratic towers (Section 3.3).

2.2 The invariant-coloring framework

We use the probabilistic formulation of Hadwiger–Nelson-type problems developed by the Polymath16 project [23], Ágoston [2], and Gwyn–Stavrianos [13]. The isometry group $E(2) = \mathrm{Isom}(\mathbb{R}^2)$ is solvable, hence amenable, so it admits a left-invariant mean Λ on $\ell^\infty(E(2))$ (this uses Hahn–Banach; it is a theorem of ZFC).

Definition 2.1. Fix any proper coloring c of $G(\mathbb{R}^2; D)$ with k colors (no measurability assumed) and fix an invariant mean Λ. For $r > 0$ define the *monochromatic-pair density*

$$p_r \;=\; p_r(c, \Lambda) \;=\; \Lambda_{g \in E(2)}\Big[\,\mathbf{1}\big(c(g \cdot (0,0)) = c(g \cdot (r,0))\big)\Big] \;\in [0,1].$$

By left-invariance, for *every* ordered pair (x, y) with $|x - y| = r$ we have $\Lambda_g[\mathbf{1}(c(gx) = c(gy))] = p_r$. This yields the transfer principle we use throughout:

Lemma 2.2 (Finite transfer)**.** *Let $Q \subset \mathbb{R}^2$ be finite with designated pairs $(a_1, b_1), \ldots, (a_m, b_m)$, $|a_i - b_i| = r_i$, and let $t_{\min}$ (resp. $t_{\max}$) be the minimum (resp. maximum), over all proper k-colorings of the finite graph $G(Q; D)$, of the number of monochromatic designated pairs. Then for every proper k-coloring c of $G(\mathbb{R}^2; D)$ and every invariant mean Λ,*

$$t_{\min} \;\leq\; \sum_{i=1}^{m} p_{r_i}(c, \Lambda) \;\leq\; t_{\max}.$$

Proof. For each $g \in E(2)$, the restriction of c to gQ is a proper k-coloring of $G(gQ; D) \cong G(Q; D)$, so the number of monochromatic designated pairs of gQ lies in $[t_{\min}, t_{\max}]$ pointwise in g. Apply Λ and use linearity together with the paragraph above, which gives $\Lambda_g[\mathbf{1}(c(ga_i) = c(gb_i))] = p_{r_i}$. □

Only one-sided bounds are needed in applications: a lower bound $t \le t_{\min}$ or an upper bound $t \ge t_{\max}$. Each bound we use is certified by refuting the decision statement "a proper k-coloring with $\le t-1$ (resp. $\ge t+1$) monochromatic designated pairs exists": the statement is encoded as a CNF formula, and its unsatisfiability is certified by a DRAT proof trace checked by `drat-trim` [26]. (The optima were located with an exact MaxSAT solver, RC2 over CaDiCaL, and are attained by explicit colorings, but only the DRAT-certified direction enters the proofs.)

2.3 Prior work used directly

De Grey's graph [8] and its successors [14, 11, 22] give $\chi(\mathbb{R}^2) \ge 5$; the 509-vertex record and its 2406-edge trim are from [22, 9]. The two-distance 6-chromatic graphs we improve are from [21, 12, 10]. The probabilistic framework is from [23, 2], and the finite-graph transfer of badness bounds from [13]. The $\chi \ge 7$ sketch is Parts' [20]. Set-theoretic sensitivity of chromatic numbers of distance graphs is due to Shelah–Soifer [24]; general background is in Soifer [25].

3 Seven colors for two distances

Throughout this section $k = 6$, $D = \{1, d\}$, and c is an arbitrary proper 6-coloring of $G(\mathbb{R}^2; \{1, d\})$, assumed to exist for contradiction; $p_r = p_r(c, \Lambda)$ for a fixed invariant mean. The two auxiliary distances are $\sqrt{2}$ and $1 + \sqrt{3}$ — not themselves forbidden, but tied to $\{1, d\}$ by the identity (1).

3.1 The V8 pigeonhole

Parts' *V8 gadget* consists of two concentric squares: the unit square $I = \{(\pm\frac{1}{2}, \pm\frac{1}{2})\}$ and the outer square $O = \{(\pm s, 0), (0, \pm s)\}$ with $s = d/\sqrt{2} = (1 + \sqrt{3})/2$.

Lemma 3.1. *$G(I \cup O; \{1, d\}) = K_8$ minus the perfect matching consisting of the four "diagonals": the two inner diagonals at distance $\sqrt{2}$ and the two outer diagonals at distance $2s = 1 + \sqrt{3}$.*

Proof. Direct computation. Inner sides have length 1; inner diagonals $\sqrt{2}$. Outer sides have squared length $2s^2 = 2 + \sqrt{3} = d^2$; outer diagonals $2s = 1 + \sqrt{3}$. For an inner corner and an outer vertex, e.g. $(\frac{1}{2}, \frac{1}{2})$ and $(s, 0)$: the difference is $(\frac{\sqrt{3}}{2}, \frac{1}{2})$ of length 1; and against $(-s, 0)$ the squared distance is $\left(\frac{2+\sqrt{3}}{2}\right)^2 + \frac{1}{4} = 2 + \sqrt{3} = d^2$. By symmetry each inner vertex is at distance 1 from the two nearest outer vertices and d from the two farthest. Thus all 24 pairs except the four diagonals are edges. □

Proposition 3.2 (V8 pigeonhole)**.** *Every proper 6-coloring of $G(I \cup O; \{1, d\})$ makes at least 2 of the four diagonal pairs monochromatic. The bound is exact, and all 11 diagonal patterns of weight ≥ 2 are realizable. Certificate: `certificates/seven-colors/v8cert.cnf` (proper 6-coloring $\wedge$ at most one monochromatic diagonal; 84 variables, 267 clauses) is UNSAT, with DRAT trace `v8cert.drat` verified by `drat-trim`.*

Proof. By Lemma 3.1 every color class is either a single vertex or one of the four diagonal pairs. If exactly j diagonals are monochromatic, the coloring uses $8 - j$ colors, so $8 - j \le 6$ forces $j \ge 2$; conversely, for any $j \ge 2$ diagonals, coloring each of them with its own color and the remaining $8 - 2j$ vertices with distinct further colors uses $8 - j \le 6$ colors. The DRAT certificate checks the first statement mechanically. □

By Lemma 2.2 with the four diagonals as designated pairs (two at $\sqrt{2}$, two at $1+\sqrt{3}$) and $t_{\min} = 2$:

Corollary 3.3. $2p_{\sqrt{2}} + 2p_{1+\sqrt{3}} \;\geq\; 2$, *i.e.* $p_{\sqrt{2}} + p_{1+\sqrt{3}} \;\geq\; 1$.

3.2 The chain configurations and the exact identity

The counterweight to Corollary 3.3 is a pair of 12-point configurations bounding $p_{\sqrt{2}}$ and $p_{1+\sqrt{3}}$ from above. Fix a radius $r \in \{\sqrt{2},\, 1+\sqrt{3}\}$ and let

$$\varphi_1 = 2\arcsin\frac{1}{2r}, \qquad \varphi_d = 2\arcsin\frac{d}{2r}$$

be the angles subtending chords of length 1 and d on a circle of radius r; equivalently $\cos\varphi_1 = 1 - \frac{1}{2r^2}$ and $\cos\varphi_d = 1 - \frac{d^2}{2r^2}$. Concretely,

$$r=\sqrt{2}: \quad \cos\varphi_1 = \tfrac{3}{4}, \quad \cos\varphi_d = \tfrac{2-\sqrt{3}}{4}; \qquad r=1+\sqrt{3}: \quad \cos\varphi_1 = \tfrac{2+\sqrt{3}}{4}, \quad \cos\varphi_d = \tfrac{3}{4},$$

a pleasing duality: the two radii exchange the roles of the two cosines.

Set $A = \varphi_d - \varphi_1$ and $B = \varphi_d + \varphi_1$, and define the *mixed chain* $X_0, \ldots, X_6$ by $X_0 = (0,0)$ and $X_{i+1} = X_i + r(\cos h_i, \sin h_i)$, where the heading sequence has period four:

$$(h_0, h_1, h_2, h_3) \;=\; \big(0,\; \pi - A,\; -(A+B),\; \pi - B\big), \qquad h_{i+4} = h_i.$$

Thus consecutive chain points are at distance exactly r. The key fact is that the four-step displacement lands *exactly* on a forbidden distance:

Lemma 3.4 (Four-step closure)**.** *For all i, $X_{i+4} - X_i$ is the same vector, of length $(d^2-1)/r$. Consequently, by (1): for $r=\sqrt{2}$, $|X_{i+4}-X_i| = (1+\sqrt{3})/\sqrt{2} = d$; for $r = 1+\sqrt{3}$, $|X_{i+4}-X_i| = 1$.*

Proof. Identifying $\mathbb{R}^2$ with $\mathbb{C}$, $X_{i+4} - X_i = r\sum_{j=i}^{i+3} e^{\mathrm{i}h_j} = r\sum_{j=0}^{3} e^{\mathrm{i}h_j}$, since the headings have period four. Now

$$\sum_{j=0}^{3} e^{\mathrm{i}h_j} \;=\; 1 - e^{-\mathrm{i}A} + e^{-\mathrm{i}(A+B)} - e^{-\mathrm{i}B} \;=\; \big(1-e^{-\mathrm{i}A}\big)\big(1-e^{-\mathrm{i}B}\big),$$

whose modulus is $4\sin\frac{A}{2}\sin\frac{B}{2} = 2\big(\cos\frac{B-A}{2} - \cos\frac{B+A}{2}\big) = 2(\cos\varphi_1 - \cos\varphi_d) = 2\cdot\frac{d^2-1}{2r^2} = \frac{d^2-1}{r^2}$. Multiplying by r gives $|X_{i+4} - X_i| = (d^2-1)/r$. □

So for $r = \sqrt{2}$ the pairs (X_i, X_{i+4}), $i = 0,1,2$, are genuine d-edges of the two-distance graph, and for $r = 1+\sqrt{3}$ they are unit edges: the chain's long-range structure is policed by the forbidden distances even though its steps are not. To stiffen the configuration we add five *apexes* $C_1, \ldots, C_5$: C_i is the rotation of X_{i-1} about X_i through $\pm\varphi_1$ (one branch per i). Then $|C_iX_i| = r$ and $|C_iX_{i-1}| = 1$ automatically, and the branch is chosen so that $|C_iX_{i+1}| = d$ — exactly, for every i, at both radii (Section 3.3); the apexes are distinct from all chain points. The resulting configuration $Q_r = \{X_0, \ldots, X_6, C_1, \ldots, C_5\}$ has 12 points; its $\{1,d\}$-graph has 14 edges (5 unit and 9 at distance d for $r=\sqrt{2}$; 9 and 5 respectively for $r = 1+\sqrt{3}$). The *charged slots* are the eleven designated pairs at distance r:

$$(X_i, X_{i+1}) \;\; (0 \leq i \leq 5) \qquad \text{and} \qquad (X_i, C_i) \;\; (1 \leq i \leq 5).$$

Proposition 3.5 (Chain caps). *For $r = \sqrt{2}$ (chain A) and for $r = 1+\sqrt{3}$ (chain B): in every proper 6-coloring of $G(Q_r; \{1, d\})$, at most 5 of the 11 charged slots are monochromatic, and 5 is attained. Certificates (in* `certificates/seven-colors/`*):* `capA.cnf` *and* `capB.cnf` *(proper 6-coloring $\wedge$ at least 6 monochromatic slots; 188 variables, 646 clauses each) are UNSAT with DRAT traces* `capA.drat`*,* `capB.drat` *verified by* `drat-trim`*; independently,* `verify_exact.py` *re-proves the cap by exhaustive enumeration of slot subsets on the exact edge lists of Section 3.3 and exhibits a coloring attaining 5.*

By Lemma 2.2 with the eleven slots (all at distance r) and $t_{\max} = 5$:

Corollary 3.6. *$11\, p_{\sqrt{2}} \le 5$ and $11\, p_{1+\sqrt{3}} \le 5$; that is, $p_{\sqrt{2}} \le \frac{5}{11}$ and $p_{1+\sqrt{3}} \le \frac{5}{11}$.*

Thus a single 12-point configuration forces slot mono-density $\le \frac{5}{11} = \frac{1}{2} - \frac{1}{22}$, and 5/11 is optimal for this configuration. Parts' sketch [20] instead obtains a bound $p_r < \frac{1}{2}$ by counting along long zigzag chains, without an explicit constant.

3.3 Exact-arithmetic certification of the chains

Every geometric statement about $Q_{\sqrt{2}}$ (chain A) and $Q_{1+\sqrt{3}}$ (chain B) — including the apex placements — has been verified in exact symbolic arithmetic; no floating point enters the $\chi \ge 7$ certificates. The verifier is `certificates/seven-colors/verify_exact.py` (∼250 lines, on `sympy` + `python-sat`); its design:

- *Towers.* All coordinates of chain A lie in $K_A = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{7}, S)$ with $S = \sqrt{9+4\sqrt{3}}$ (note $\sin\varphi_d = S/4$, $\sin\varphi_1 = \sqrt{7}/4$ at $r = \sqrt{2}$), a tower of nominal degree 16; those of chain B lie in $K_B = \mathbb{Q}(\sqrt{3}, \sqrt{7}, T)$ with $T = \sqrt{9-4\sqrt{3}}$ (here $\sin\varphi_1 = T/4$, $\sin\varphi_d = \sqrt{7}/4$), of degree 8. Elements are polynomials in the generators, reduced to normal form modulo the defining relations $S^2 = 9+4\sqrt{3}$, $w_2^2 = 2$, $w_7^2 = 7$, $w_3^2 = 3$ (resp. $T^2 = 9-4\sqrt{3}$, …). Under a lexicographic order the relation set has pairwise-coprime pure-square leading monomials, hence is a Gröbner basis, so normal forms are unique and the quotient ring has dimension 16 (resp. 8).

- *Soundness certificate.* Nonzero normal form certifies a nonzero field element only if the quotient ring *is* the field. This is certified separately: the minimal polynomial of the primitive element $\sqrt{2}+\sqrt{3}+\sqrt{7}+\sqrt{9+4\sqrt{3}}$ (resp. $\sqrt{3}+\sqrt{7}+\sqrt{9-4\sqrt{3}}$) is computed and its degree confirmed to be 16 (resp. 8). Every subsequent equality/inequality decision is therefore exact and unconditional.

- *Checks.* For each chain: the headings are exact unit vectors; all six steps satisfy $|X_iX_{i+1}|^2 = r^2$ exactly; the five apexes satisfy $|C_iX_i|^2 = r^2$, $|C_iX_{i-1}|^2 = 1$, and $|C_iX_{i+1}|^2 = d^2 = 2+\sqrt{3}$ exactly; the three long pairs satisfy $|X_iX_{i+4}|^2 = (d^2-1)^2/r^2$ exactly ($= 2+\sqrt{3}$ for chain A, $= 1$ for chain B), confirming Lemma 3.4 independently; and *all* 66 vertex pairs are classified exactly, reproducing the 14-edge sets used by the SAT certificates and confirming that all 11 charged slots are exactly at distance r and are non-edges. The exact coordinates agree with the floating-point coordinate files (`chain12.json`, `chain12b.json`) to 10^{-9}.

- *Caps on exact edges.* Proposition 3.5 is re-proved on the exact edge lists by exhaustive enumeration: all $\binom{11}{6} = 462$ six-subsets of the slots are unsatisfiable as simultaneous monochromatic sets under proper 6-coloring, while a five-subset is satisfiable; hence the maximum is exactly 5.

3.4 Proof of Theorem 1.1

Proof of Theorem 1.1. Suppose c is a proper 6-coloring of $G(\mathbb{R}^2;\{1,d\})$ and fix an invariant mean Λ (ZFC). By Corollary 3.3, $p_{\sqrt{2}} + p_{1+\sqrt{3}} \geq 1$; by Corollary 3.6, $p_{\sqrt{2}} + p_{1+\sqrt{3}} \leq \frac{10}{11}$. Contradiction. Hence no proper 6-coloring exists and $\chi\big(\mathbb{R}^2;\{1,d\}\big) \geq 7$. □

3.5 The framework caveat

Two caveats separate Theorem 1.1 from a classical finite-witness bound such as de Grey's $\chi(\mathbb{R}^2) \geq 5$. (A third — floating-point placement of the chain apexes — was eliminated by the exact-arithmetic certification of Section 3.3: the V8 and both chains are now verified exactly, end to end.)

(i) *Nonconstructive averaging.* The densities p_r exist by an invariant mean on the amenable group $E(2)$, whose existence uses Hahn–Banach. This is a ZFC theorem, so Theorem 1.1 rules out *arbitrary* (not merely measurable) 6-colorings in ZFC. But in ZF+DC the mean need not exist, and Shelah–Soifer [24] showed chromatic numbers of distance graphs can genuinely depend on the ambient set theory. The choice-free content of this section is exactly its three finite certificates.

(ii) *No explicit finite witness yet.* By the de Bruijn–Erdős theorem a finite 7-chromatic $\{1,d\}$-graph exists, but the argument does not exhibit one. Section 4 makes the gap precise and reduces it to a combinatorial design problem; Section 4.2 reports computational evidence that the most obvious design vocabulary does not suffice.

4 Toward a fully finite witness

4.1 The counting blueprint

The two halves of the proof are pointwise statements about finite configurations; only the step from "every placement" to "density" uses the invariant mean. A single finite graph would suffice if the slots of many chain copies could be made to *coincide* with the diagonals of many V8 copies:

Proposition 4.1 (Uniform-multiplicity blueprint)**.** *Suppose a finite point set W contains N V8 copies and a family $\mathcal{C}$ of chain copies (of types A and B) such that the multiset of charged slots of $\mathcal{C}$ covers the $4N$ diagonal pairs of the V8s, each with the same multiplicity $m \geq 1$, and contains no other pairs. Then $G(W;\{1,d\})$ is not 6-colorable.*

Proof. Let M be the number of monochromatic diagonals in a putative proper 6-coloring. Summing Proposition 3.2 over the N V8s gives $M \geq 2N$. Each chain contributes 11 slots of which at most 5 are monochromatic (Proposition 3.5); summing over $\mathcal{C}$ and using the uniform multiplicity, $mM \leq \frac{5}{11} \cdot 4Nm$, i.e. $M \leq \frac{20}{11}N < 2N$. Contradiction. □

The design problem is thus to realize such an incidence structure with actual plane geometry — every slot at distance $\sqrt{2}$ or $1+\sqrt{3}$ being simultaneously a diagonal of some unit or outer square whose V8 completion is present. Exact uniform multiplicity is a strong requirement; a defect-tolerant weighted version of Proposition 4.1, allowing some slots to be uncovered or covered unevenly, appears feasible, but we have not written out a complete proof and make no claim about it here. A naive faithful closure of the required incidences is Følner-type and very large, which motivates:

Question 4.2 (Gatekeeper question)**.** Does a finite $\{1, d\}$-graph exist containing two designated r-pairs in a rational-angle relative position such that no proper 6-coloring makes both pairs monochromatic? The natural target is "not both diagonals of one unit square monochromatic" (90°, self-supplied by V8s).

A positive answer would allow far smaller finite witnesses than a naive closure. *Computational observation* (not certified): we searched closures of the ζ_{24} lattice (built from a 12-fold wheel of De Neve-style gadgets about a hub, whose output ring is a regular 12-gon with side exactly $\sqrt{2}$ and 60° chords exactly $1+\sqrt{3}$), including neighborhoods of a designated unit square grown for eight rounds inside a 24,881-point universe, up to 1837 points and 16,116 edges. In all of them, 6-colorings making both diagonals of the designated square monochromatic existed and were found quickly. Within this horizon we found no rational-angle cap gadget.

4.2 Computational observations

The following were obtained by exhaustive SAT enumeration and exact symbolic computation. The code and logs are not part of the certificate bundle and the arguments have not been written up, so we report them as observations that guided the search, not as theorems.

- *Duplication and trees.* Two copies of a configuration sharing slots contributed at most one useful clause in all cases tested, and tree-like slot incidence structures never forced a monochromatic pair.
- *No short angle relations.* No integer relation of weight ≤ 7 was found among $\alpha = \arccos\frac{2+\sqrt{3}}{4}$, $\beta = \arccos\frac{3}{4}$, $\gamma = \arccos\frac{2-\sqrt{3}}{4}$, $\delta' = \arccos\frac{3\sqrt{6}-\sqrt{2}}{8}$, and 30°, so there are no short "odd spindle" cycles among the relevant angles.
- *Two-V8 sharing.* For two V8s sharing 4 points, all $121 = 11^2$ joint weight-≥ 2 diagonal patterns were feasible, so pattern-level reasoning over such pairs of V8s forces nothing.
- *Pattern sets.* The sets of achievable monochromatic-slot patterns were enumerated for the V8 (all patterns of weight ≥ 2, as in Proposition 3.2) and for both chains. Restricting chain A to a 7-slot subfamily gives cap $3/7$, the lowest cap ratio among all 2047 nonempty slot subsets.
- *Coincidences.* Coincidence scans over the hinge families (A–A, B–B, and A–B at 45°) found that two chain placements share at most one charged slot, and the translation modules governing periodic ladder assemblies had full $\mathbb{Z}$-rank in exact arithmetic over $\mathbb{Q}(\sqrt{3}, \sqrt{7})$ and the towers K_A, K_B of Section 3.3. This suggests that slot multiplicity cannot be accumulated cheaply by crossing chains.
- *Linear-programming duals.* A Farkas-dual linear program over the enumerated pattern sets, with the coincidence constraints above, was infeasible for the inventory consisting of V8s, chains and trimmed chains, suggesting that these gadgets alone do not yield a finite contradiction.

If these observations are correct, a finite witness needs further ingredients, for example *generic-position pigeonhole gadgets* (V8-style pigeonholes operating in generic relative position, which is also the direction of Question 4.2) or *replacement cap gadgets at new radii*, such as the star-polygon family of Section 7.

5 The record six-chromatic graphs

5.1 A 26-vertex 6-chromatic $\{1, d\}$-graph

Theorem 5.1. *The* 26 *points of Table 1 span* 85 *pairs at distance exactly* 1 *(*48 *pairs) or exactly* d *(*37 *pairs), and the graph on these pairs has chromatic number* 6 *and is vertex-critical. Hence* $\chi\big(\mathbb{R}^2; \{1, d\}\big) \geq 6$ *via a* 26*-vertex witness — to our knowledge the smallest* 6*-chromatic two-distance graph known for any ratio. Certificates:* `certificates/g26/`.

Each vertex is an integer 8-vector $[a_0, \ldots, a_7]$ encoding the point

$$\Big(\tfrac{a_0+a_1\sqrt{3}+a_2\sqrt{7}+a_3\sqrt{21}}{8},\ \tfrac{a_4+a_5\sqrt{3}+a_6\sqrt{7}+a_7\sqrt{21}}{8}\Big) \in \mathbb{Q}(\sqrt{3}, \sqrt{7})^2.$$

Distances are verified in exact integer arithmetic: writing a coordinate difference as (p, q, r, s) in the basis $(1, \sqrt{3}, \sqrt{7}, \sqrt{21})/8$, its square contributes

$$\big(p^2 + 3q^2 + 7r^2 + 21s^2,\ 2pq + 14rs,\ 2pr + 6qs,\ 2ps + 2qr\big)$$

in the basis $(1, \sqrt{3}, \sqrt{7}, \sqrt{21})/64$; summing the x and y contributions, a pair is a unit edge iff the total is $(64, 0, 0, 0)$ and a d-edge iff it is $(128, 64, 0, 0)$, i.e. squared distance $2 + \sqrt{3} = d^2$.

Vertices of the 26-vertex graph $([a_0, a_1, a_2, a_3 \mid a_4, a_5, a_6, a_7])$	
$[-11, 0, 0, 1 \mid 0, 3, 1, 0]$	$[-3, 3, 1, 1 \mid 3, 3, 1, -1]$
$[-8, -1, 1, 0 \mid -1, 0, 0, -1]$	$[-3, 4, 0, 1 \mid -4, 3, 1, 0]$
$[-8, 0, 0, 0 \mid 0, 0, 0, 0]$	$[-3, 4, 0, 1 \mid 4, 3, 1, 0]$
$[-8, 3, -1, 0 \mid -3, 0, 0, -1]$	$[-3, 4, 2, 1 \mid 2, 3, 1, 0]$
$[-8, 3, 1, 0 \mid -5, 0, 0, -1]$	$[-2, 3, -1, 0 \mid -3, 0, -2, -1]$
$[-8, 3, 1, 0 \mid 3, 0, 0, -1]$	$[0, 3, -1, 0 \mid -3, 0, 0, -1]$
$[-7, 0, 0, 1 \mid 0, -1, 1, 0]$	$[0, 3, 1, 0 \mid -5, 0, 0, -1]$
$[-7, 4, 0, 1 \mid -4, -1, 1, 0]$	$[0, 3, 1, 0 \mid 3, 0, 0, -1]$
$[-6, 0, 0, 0 \mid 0, 0, 2, 0]$	$[0, 4, 0, 0 \mid -4, 0, 0, 0]$
$[-6, 1, 1, 0 \mid -1, 0, 2, 1]$	$[0, 7, 1, 0 \mid -1, 0, 0, -1]$
$[-6, 3, -1, 0 \mid -3, 0, 2, -1]$	$[1, 4, 0, 1 \mid -4, -1, 1, 0]$
$[-6, 4, 0, 0 \mid -4, 0, 2, 0]$	
$[-5, 3, 1, 1 \mid 3, 3, -1, -1]$	
$[-4, 3, 1, 0 \mid 3, 4, 0, -1]$	
$[-3, 0, 0, 1 \mid 0, 3, 1, 0]$	

Table 1: Integer coordinate vectors of the 26-vertex 6-chromatic $\{1, (\sqrt{6}+\sqrt{2})/2\}$-graph. Point encoded by $\big((a_0 + a_1\sqrt{3} + a_2\sqrt{7} + a_3\sqrt{21})/8,\ (a_4 + a_5\sqrt{3} + a_6\sqrt{7} + a_7\sqrt{21})/8\big)$.

5.2 A 30-vertex 6-chromatic $\{1, \sqrt{3}\}$-graph

Theorem 5.2. *The* 30 *points of Table 2 span* 100 *pairs at distance exactly* 1 *(*72*) or exactly* $\sqrt{3}$ *(*28*), and the graph on these pairs has chromatic number* 6 *and is vertex-critical. Hence* $\chi\big(\mathbb{R}^2; \{1, \sqrt{3}\}\big) \geq 6$ *via a* 30*-vertex witness (previously* 33 *[10]). Certificates:* `certificates/g30/`.

Here each vertex is an integer 4-vector $[a_0, a_1, a_2, a_3]$ encoding

$$\Big(\tfrac{a_0+a_1\sqrt{5}}{8},\ \tfrac{a_2\sqrt{3}+a_3\sqrt{15}}{8}\Big) \in \mathbb{Q}(\sqrt{3}, \sqrt{5})^2, \qquad a_0 \equiv a_1,\ a_2 \equiv a_3 \pmod 2.$$

For a difference vector (p, q, r, s), $64 \cdot \mathrm{dist}^2 = p^2 + 5q^2 + 3r^2 + 15s^2 + (2pq + 6rs)\sqrt{5}$, so a unit edge means exactly $(64, 0)$ and a $\sqrt{3}$-edge exactly $(192, 0)$.

Vertices of the 30-vertex graph ($[a_0, a_1, a_2, a_3]$)		
$[-6, 0, -2, 0]$	$[-1, -3, -1, 1]$	$[1, 3, -1, 1]$
$[-5, -3, -1, 1]$	$[-1, -3, 1, 3]$	$[1, 3, 3, 1]$
$[-5, 3, -3, 1]$	$[-1, 3, -1, 3]$	$[2, 0, -2, 0]$
$[-4, 0, -2, 2]$	$[-1, 3, 1, 1]$	$[2, 0, 0, 2]$
$[-4, 0, 0, 0]$	$[-1, 3, 3, 3]$	$[2, 0, 2, 0]$
$[-4, 0, 0, 4]$	$[0, 0, -2, 2]$	$[3, -3, 1, 3]$
$[-3, -3, -1, 3]$	$[0, 0, 0, 0]$	$[3, 3, -1, 3]$
$[-3, -3, 1, 1]$	$[0, 0, 2, 2]$	$[4, 0, 0, 4]$
$[-3, 3, -1, 1]$	$[1, -3, -3, 1]$	
$[-2, 0, -2, 4]$	$[1, -3, 1, 1]$	
$[-2, 0, 0, 2]$		
$[-1, -3, -3, 3]$		

Table 2: Integer coordinate vectors of the 30-vertex 6-chromatic $\{1, \sqrt{3}\}$-graph. Point encoded by $\big((a_0 + a_1\sqrt{5})/8,\ (a_2\sqrt{3} + a_3\sqrt{15})/8\big)$.

5.3 Method: SAT-guided iterated local search over module closures

Both graphs were found by the same pipeline, seeded from the published coordinate tables of De Neve et al. [10] (their 33-vertex $\{1, \sqrt{3}\}$-graph and 117-vertex $\{1, d\}$-graph).

- *Universe.* A bounded-coefficient slice of the $\mathbb{Z}$-module closure of the seed coordinates: all points $p + q - r$ over the current point set staying in the same coset of the coordinate module (for $\{1, \sqrt{3}\}$: $\{((a_0 + a_1\sqrt{5})/8, (a_2\sqrt{3} + a_3\sqrt{15})/8) : a_i \in \mathbb{Z},\ a_0 \equiv a_1,\ a_2 \equiv a_3 \pmod 2\}$, $\approx 10^4$ points within radius 3.4). The closure contains coefficient patterns the seed constructions never generate; the 26-vertex graph uses several such points.
- *Oracle.* An incremental CaDiCaL instance with per-vertex selector assumptions answers "is this subset still 6-chromatic?" in microseconds.
- *Search.* Randomized greedy vertex deletion descends to critical subgraphs; perturbation kicks (adjacency-weighted candidate insertion, remove-k, re-descend) escape local minima. The 30-vertex minimum was reached within $\sim 10^4$ oracle calls ($\approx$ 1 minute on an Apple M4 Pro); it shares 23 vertices with the 33-vertex seed, omits 10, and adds 7 new. In the $\{1, d\}$ search the descent first stopped at 29 vertices because the clique used for symmetry breaking had been exempted from deletion; a final pass without this exemption reduced the graph to 26 vertices.

5.4 Verification recipe

Every claim in Theorems 5.1 and 5.2 is re-derivable from the integer vector tables alone; the files are in `certificates/g26/` and `certificates/g30/`.

1. *Exact edges.* Recompute all pairwise squared distances in integer arithmetic as above; no floating point enters any certificate.

2. *Non-5-colorability.* The direct 130– resp. 150-variable CNF (one variable per vertex-color pair, no symmetry-breaking assumptions; `record26.cnf`, `record30.cnf`) is UNSAT, with DRAT traces `record26.drat` (18,518 lines) and `record30.drat` (compressed as `record30.drat.xz`) checked by `drat-trim` (`s VERIFIED`).

3. 6-*colorability and criticality.* Explicit 6-colorings exist, and all single-vertex-deleted subgraphs are 5-colorable; both are found by SAT in the verifiers.

4. *Standalone verifiers.* `verify117best.py deneve117_min.json` and `verify30.py` (~ 60 lines each; dependency: `python-sat`) recompute the edge sets exactly from the integer vectors and re-solve non-5-colorability (for the 26-vertex graph with two different SAT solvers), 6-colorability and vertex-criticality from scratch in seconds.

Remark 5.3. Whether a 25-vertex (or smaller) 6-chromatic two-distance graph exists for any ratio is open. Note that the bound $v_6 \geq 62$ for unit-distance graphs (Theorem 6.3) does not apply to two-distance graphs, where only trivial lower bounds are known for $\{1, \sqrt{3}\}$.

6 Unit-distance results

6.1 $e_5 \leq 2257$

The smallest known 5-chromatic unit-distance graph has 509 vertices and 2442 edges [22, 9]. It was known not to be edge-critical: de Grey and Parts removed 36 edges non-exhaustively, giving $e_5 \leq 2406$, and noted room for improvement [9].

Theorem 6.1. *The* 509*-vertex graph contains a spanning subgraph with* 2257 *edges (*185 *removed) that is not* 4*-colorable. Hence* $e_5 \leq 2257$*.*

The subgraph is listed in `certificates/e5-2257/edge2257_graph.json` (an edge list on the vertex numbering $0, \ldots, 508$ of the 509-vertex graph). Its non-4-colorability is certified by `edge2257.cnf`/`edge2257.drat` (compressed with `xz`; binary DRAT), checked by `drat-trim` in about thirty seconds. The CNF has one at-least-one-color clause per vertex, one clause per edge and color, and three unit clauses fixing the colors of the three vertices of one triangle of the subgraph, which is a valid symmetry breaking. The bundle also contains the exact algebraic coordinates of the 509 vertices (`parts509_coordinates.vtx`, from [22]) and a script `check_unit.py` which confirms, at 60-digit precision, that these points have exactly 2442 unit-distance pairs, with no near-unit pair, and that the 2257 listed edges are among them.

The first trim used the same incremental selector-assumption technique as Section 5.3, over edges instead of vertices: repeatedly attempt to delete a random edge, keeping the deletion iff the remainder is still not 4-colorable, until no edge is removable. Such greedy runs end at 2278–2285 edges depending on the order. The 2257-edge subgraph comes from a decomposition. Split the vertices as $L = \{0, \ldots, 373\}$ and $S = \{374, \ldots, 508\}$, which are joined by only 30 edges. Each candidate L-part realizes a set of 4-colorings on the interface; a combination is non-4-colorable exactly when no interface coloring extends to both sides. We re-minimized the union of two dissimilar L-parts with the S-part frozen, which gave 1717 edges inside L, and ran an iterated local search on the S-side subproblem, which gave 540 edges inside S or across the cut. The certificate does not depend on this reasoning: the DRAT proof certifies the combined graph directly. To speed up the searches we used a tabu-search 4-coloring heuristic, which certifies

that an edge is necessary by exhibiting a 4-coloring of the graph minus that edge; this replaces most SAT calls. *Computational observation* (not part of the certificate bundle): the 2257-edge subgraph is edge-critical. For 1186 of its edges a tabu-search 4-coloring of the graph minus that edge was found; for the other 1071 edges, removing the edge leaves a graph that CaDiCaL finds 4-colorable. Further progress therefore needs non-greedy moves. The true e_5 may be lower still.

6.2 $e_6 \geq 322$ and $v_6 \geq 62$

Let e_k and v_k denote the least numbers of edges and of vertices of a k-chromatic unit-distance graph in the plane. Lower bounds come from colorings that are *almost* proper. For a measurable coloring c of the plane that is periodic with respect to a lattice Λ, let

$$p(c) = \Pr\big[c(x) = c(x+u)\big],$$

where x is uniform in a fundamental domain of Λ and u is uniform on the unit circle.

Lemma 6.2 (Gwyn–Stavrianos [13]; de Grey–Parts [9])**.** *If c is a periodic measurable $(k-1)$-coloring of the plane and G is a unit-distance graph with $e < 1/p(c)$ edges, then G is $(k-1)$-colorable. Hence $e_k \geq \lceil 1/p(c) \rceil$.*

Proof. Place G by a uniformly random rigid motion (a uniform translation modulo Λ composed with a uniform rotation). Each edge is then a random unit segment, so it is monochromatic with probability $p(c)$, and the expected number of monochromatic edges is $e\,p(c) < 1$. Since this number is a non-negative integer, it is 0 with positive probability. Almost every placement puts no vertex on the null set of cell boundaries, so some placement does both, and restricting c to the placed vertices is a proper $(k-1)$-coloring of G. □

De Grey and Parts obtained $e_5 \geq 99$ from Croft's tiling and the regular hexagonal 4-coloring, and $e_6 \geq 182$ by superimposing a fifth color on a 4-coloring at a random offset [9]. We optimize all five colors jointly.

Theorem 6.3. *There is a 5-coloring c_5 of the plane, periodic with respect to a hexagonal lattice and composed of 4230 convex polygons per period with rational vertices, such that*

$$p(c_5) \leq 0.0031082241, \qquad 1/p(c_5) \geq 321.727.$$

Consequently every 6-chromatic unit-distance graph has at least 322 edges and at least 59 vertices.

Proof. The bound on $p(c_5)$ is certified as described below. Then $e_6 \geq 322$ follows from Lemma 6.2. For the vertex bound, Ágoston and Pálvölgyi [1, Prop. 12] show that a unit-distance graph with n vertices and e edges satisfies $n \geq 2r - 11 + 24/n + (1-s)\binom{\lfloor r \rfloor}{2} + s\binom{\lceil r \rceil}{2}$, where $r = 2e/n$ and $s = r - \lfloor r \rfloor$. With $n = 61$ this forces $e \leq 321 < 322$. □

The coloring is shown in Figure 1. Four colors form roughly convex blobs of diameter just under 1, each of density about 0.225. This is close to Croft's density 0.2293 for a set avoiding distance 1. The fifth color, of density 0.097, fills a connected network of channels between the blobs. The period lattice is spanned by $(7, 0)$ and $(7/2, 6.0622)$, a 7×7 hexagonal supercell; other periods we tried were markedly worse.

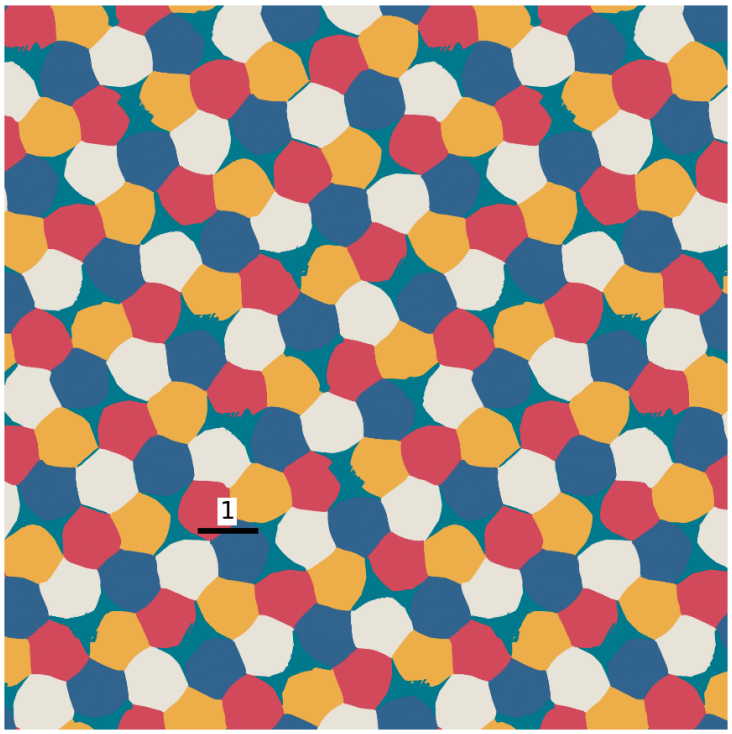


Figure 1: The periodic 5-coloring c_5 of Theorem 6.3, shown over several periods. A random unit segment (bar) is monochromatic with probability 0.0031081.

Certificate. The file `certificates/e6-lower-bound/c5_best.json` lists the period basis and the 4230 convex pieces in exact rational coordinates. The verifier `certify.py` works in two stages.

(i) In exact rational arithmetic, it checks that every cell is strictly convex, that lattice translates of the cells have pairwise disjoint interiors (by a separating-axis test), and that the cell areas sum to the covolume. Together these show that the translates tile the plane.

(ii) Write P_i for the cells. Then $p = \frac{1}{2\pi \operatorname{covol}} \sum_{i,j} \sum_{\lambda} \int_0^{2\pi} \operatorname{area}\big(P_i \cap (P_j + \lambda - u_\theta)\big)\, d\theta$, where the sum runs over pairs of cells of the same color and lattice vectors λ. The verifier encloses each integral in arb ball arithmetic. It computes every angle at which a vertex of one polygon crosses an edge line of the other. Between consecutive crossing angles the intersection area is a trigonometric polynomial of degree 2 in θ. The verifier fits it exactly from 5 rigorous evaluations, checks the fit at a sixth, and integrates in closed form. Small balls around the crossing angles are bounded crudely by the cell areas.

The run takes about 15 minutes with three worker processes and returns $p \in [0.003108 \pm 2.3 \cdot 10^{-7}]$, with upper endpoint 0.00310822410. As an independent check, a separate short C program (`pmono.c`) clips polygons in floating point and integrates over θ by the midpoint rule. It gives $p = 0.003108121$ at 1000 and 4000 nodes, so $1/p = 321.738$. A Monte Carlo estimate on an earlier version of the coloring also agreed. For calibration, the same verifier applied to the regular hexagonal 4-coloring at the optimal diameter 1.1335 returns $1/p_4 = 98.4956$, which reproduces $e_5 \geq 99$ and $v_5 \geq 28$ [9].

Search. We first annealed pixel colorings on a GPU, using an exact kernel for genuinely pixelated colorings, and upsampled the best ones repeatedly. We then fitted a power diagram to the pixel map and optimized its cell vertices with L-BFGS, using an analytic gradient of p; this reached $1/p \approx 300.7$. Next we converted the power diagram into a periodic planar map with shared vertices. Optimizing those vertex positions directly keeps the tiling exact and allows non-convex color regions. Alternating rounds of L-BFGS with merging of short boundary edges reached $1/p \approx 310.3$. There the optimizer stalls at thin needle-shaped tips of cells, which are about to self-intersect. We cut each tip with interior angle below $12°$: the thin triangle at the tip is reassigned to the neighboring region across one of its edges. This keeps the map exact and barely changes p, but it removes the barrier. Further rounds reached $1/p \approx 315.3$. The final coloring was obtained differently. We traced the color-region boundaries of the best pixel coloring directly into a periodic planar map, simplifying each shared boundary chain with Douglas–Peucker at tolerance 0.006, which gave $1/p \approx 315.4$ before any optimization. Eight rounds of the same loop (L-BFGS, merging short edges, cutting tips, subdividing long edges) then reached $1/p \approx 321.7$. Finally the vertices were rounded to rationals with denominator 10^8, and each region was cut into convex pieces by exact ear clipping followed by greedy re-merging. The resulting coloring was then certified. The following are computational observations and are not certified. Pixel colorings at resolution 0.0055 reached $1/p \approx 317.6$ in our searches, and pixel optima at the coarser resolutions 0.022 and 0.011 reached 281 and 307. The next step, $v_6 \geq 63$, needs $e_6 \geq 330$, i.e. $1/p > 329$, which appears to require a better pixel optimum than this 7×7 topology provides. For $k = 5$ we found no improvement over the hexagonal coloring. It is a stationary point, since reflection in each edge swaps the two colors there. Of 58 random-restart searches over 4-colorings, 57 converged to it.

6.3 Forced structure in 5-colorings of the plane

The transfer principle of Section 2.2 applies equally with $k = 5$ and $D = \{1\}$: a non-5-colorable $\{1, d'\}$-graph whose m d'-edges cannot all be avoided forces monochromatic pairs at distance d' in *every* proper unit-distance 5-coloring of the plane (cf. Gwyn–Stavrianos [13]). Since $\chi(\mathbb{R}^2) \geq 6$ [17], the following corollary is vacuous; we record it because its finite content — the ratio t/m of forced to designated pairs in an explicit point set — was the original motivation for the search for small graphs, and because it is the quantity in the win condition of Section 7.

Corollary 6.4. *In every proper* 5*-coloring of* $G(\mathbb{R}^2; \{1\})$*, with* p_r *the invariant densities of Definition 2.1:*

$$p_{\sqrt{3}} \;\geq\; \tfrac{1}{28}, \qquad p_d \;\geq\; \tfrac{1}{37}, \qquad p_{1/\sqrt{3}} \;\geq\; \tfrac{1}{67}.$$

Proof. Consider the 30-point set of Theorem 5.2. Any 5-coloring of the plane proper at distance 1 restricts to a coloring of the 30 points proper on the 72 unit edges; since the full graph is not 5-colorable (`g30/`), at least one of the 28 $\sqrt{3}$-pairs is monochromatic. By Lemma 2.2, $28\,p_{\sqrt{3}} \geq 1$. The same argument on the 26-point set (Theorem 5.1, `g26/`; 37 d-pairs) gives $37\,p_d \geq 1$.

For the third bound, `certificates/forced-density/` contains a set S of 67 of the 72 unit pairs of the same 30-point set such that the graph formed by all 28 $\sqrt{3}$-pairs together with S is not 5-colorable (`record67.cnf`/`record67.drat`, checked by `drat-trim`; `verify67.py` also recomputes all distances exactly and checks that S is minimal). Scale the point set by $1/\sqrt{3}$: the $\sqrt{3}$-pairs become unit pairs and the pairs of S become pairs at distance $1/\sqrt{3}$. For every placement, a coloring of the plane proper at distance 1 is proper on the former $\sqrt{3}$-pairs, so at least one pair of S is monochromatic, and Lemma 2.2 gives $67\,p_{1/\sqrt{3}} \geq 1$. □

For comparison, Parts' 397-vertex $\{1,2\}$-graph [21] implicitly forces p_2 to be of order $1/550$ in the same sense. Conversely $p_{d'} \le \frac{1}{2}$ for all $d' \ge \frac{1}{2}$ by a color-count-free argument: if $|xa| = |xb| = d'$ and $|ab| = 1$, at most one of the pairs (x,a), (x,b) is monochromatic. So any construction with $t/m > \frac{1}{2}$ would have proved $\chi(\mathbb{R}^2) \ge 6$, and Section 7 lowers this threshold to $\frac{1}{3}$ at the distance $1/\sqrt{3}$.

Remark 6.5 (Density ledger; computational observation). We also assembled about 143,000 linear inequalities among monochromatic-pair densities, each obtained from a small unit-distance configuration as in Lemma 2.2 with MaxSAT-computed bounds, and solved the resulting linear program in floating point. It returned the bound $p_{\sqrt{3}} \ge 1/12$ for invariant proper 5-colorings. The inequalities were not individually certified and the linear program was not re-solved in exact arithmetic, so we do not claim this bound (which is in any case vacuous now).

Remark 6.6 (A single-distance sweep; computational observation). For Heule's 803-vertex 5-chromatic unit-distance graph (April 2018, from the Polymath16 minimization effort [23, 14]), we tested, for each of the 428 distance classes with at least 40 point pairs, whether some proper 5-coloring avoids monochromatic pairs in that class entirely. The answer was yes for every class, so no single distance class is forced monochromatic in this graph. All classes were resolved within seconds except the class of distance 2, which took 3.4 hours of solver time before a witness coloring was found.

7 Strict caps for five colors and a win condition

(This section was written before [17] proved $\chi(\mathbb{R}^2) \ge 6$, which makes the win condition of Theorem 7.5 moot. We keep it for the cap statements, which remain valid.)

Corollary 3.6 caps a density strictly below $\frac{1}{2}$ for 6-colorings. This section develops the analogous theory on the unit-distance side: caps valid for colorings with *any* number of colors, a bipartiteness obstruction explaining where such pairwise caps cannot exist, and the resulting win condition for $\chi(\mathbb{R}^2) \ge 6$. Throughout, "proper" means proper at distance 1, and p_r is the invariant density of Definition 2.1 for such a coloring.

7.1 Color-count-free finite caps

Proposition 7.1 (Triangle and circumcenter)**.** *Let Q consist of a unit equilateral triangle $v_1v_2v_3$ and its circumcenter o, with $|o-v_i| = 1/\sqrt{3}$. In every proper coloring (any number of colors), at most one of the three pairs (o,v_i) is monochromatic. Hence $p_{1/\sqrt{3}} \le \frac{1}{3}$ for every proper coloring of the plane.*

Proof. The v_i are pairwise at distance 1, so they receive three distinct colors; o can agree with at most one of them. Transfer (Lemma 2.2) gives $3\,p_{1/\sqrt{3}} \le 1$. □

Proposition 7.2 (Unit pentagon)**.** *Let $\tau = (1+\sqrt{5})/2$. The regular pentagon with unit sides has diagonals of length τ; in every proper coloring, at most two of the five diagonals are monochromatic, so $p_\tau \le \frac{2}{5}$. Dually, the regular pentagon with unit diagonals has sides of length $1/\tau$ and caps $p_{1/\tau} \le \frac{2}{5}$.*

Proof. Label the vertices $v_0,\dots,v_4$ cyclically. If two diagonals sharing a vertex, say (v_0,v_2) and (v_2,v_4), were both monochromatic, then $c(v_0) = c(v_4)$, contradicting the unit side (v_4,v_0). So monochromatic diagonals form an independent set in the diagonal 5-cycle $v_0v_2v_4v_1v_3$, of size at most 2. The dual statement is identical with sides and diagonals exchanged. □

Proposition 7.3 (Odd stars and flowers)**.** *For odd n and $1 \le a < n/2$ with $\gcd(a,n) = 1$: place a regular n-gon so that its step-$2a$ chords have length 1; its step-a chords then have length $r = 1/(2\cos(a\pi/n))$, and in every proper coloring at most $(n-1)/2$ of the n step-a chords are monochromatic: $p_r \le \frac{n-1}{2n} < \frac{1}{2}$. Similarly the "flower" (regular m-gon with unit sides plus its center, m odd) caps $p_r \le \frac{m-1}{2m}$ at the circumradius $r = 1/(2\sin(\pi/m))$; $m = 3$ recovers Proposition 7.1. The capped radii $1/(2\cos(a\pi/n))$ are dense in $(\frac{1}{2}, \infty)$.*

Proof. If the circumradius is R, the step-k chords have length $2R\sin(k\pi/n)$; with $2R\sin(2a\pi/n) = 1$ the step-a chords have length $\sin(a\pi/n)/\sin(2a\pi/n) = 1/(2\cos(a\pi/n))$. Two step-$a$ chords sharing a vertex, (v_i, v_{i+a}) and (v_{i+a}, v_{i+2a}), cannot both be monochromatic: that would force $c(v_i) = c(v_{i+2a})$ across the unit step-$2a$ chord. So monochromatic step-a chords are independent in the step-a cycle, which is a single *odd* cycle since $\gcd(a,n) = 1$ and n is odd; an odd n-cycle has independence number $(n-1)/2$, and Lemma 2.2 gives the cap. For the flower, consecutive spokes (o, v_i), (o, v_{i+1}) cannot both be monochromatic because $|v_i v_{i+1}| = 1$, and the m spokes form an odd m-cycle under this relation. Finally, the fractions a/n with n odd, $\gcd(a,n) = 1$ and $a < n/2$ are dense in $(0, \frac{1}{2})$, so the radii $1/(2\cos(a\pi/n))$ are dense in $(\frac{1}{2}, \infty)$. □

These caps hold for arbitrary proper colorings, with no bound on the number of colors; they are elementary, and we do not claim priority for them.

7.2 A bipartiteness obstruction at $\sqrt{3}$ and 2

Why do the caps above live at $1/\sqrt{3}$, τ, $1/(2\cos(a\pi/n))$ and not at $\sqrt{3}$ and 2, the second distances of the $\{1, \sqrt{3}\}$- and $\{1, 2\}$-constructions? All of them use the same mechanism. For $r \ge \frac{1}{2}$, call an unordered pair of points at distance r an *r-slot*, and say that two r-slots $\{x, a\}$, $\{x, b\}$ sharing an endpoint *conflict* if $|a - b| = 1$. Two conflicting slots cannot both be monochromatic in a proper coloring, since that would give $c(a) = c(x) = c(b)$. Hence, for a finite family F of r-slots with conflict graph $\Gamma(F)$ (vertices the slots, edges the conflicting pairs), the monochromatic slots of any proper coloring form an independent set of $\Gamma(F)$, and Lemma 2.2 gives the *pairwise cap*

$$p_r \;\le\; \frac{\alpha(\Gamma(F))}{|F|}.$$

Propositions 7.1–7.3 are all of this form, with $\Gamma(F)$ an odd cycle. The angle at the shared endpoint of two conflicting slots is $\varphi_1(r) = 2\arcsin\frac{1}{2r}$, with $\cos\varphi_1 = 1 - \frac{1}{2r^2}$ (at $r = 1/\sqrt{3}$: $\cos\varphi_1 = -\frac{1}{2}$, $\varphi_1 = 2\pi/3$; at $r = \tau$: $\cos\varphi_1 = \tau/2$, $\varphi_1 = \pi/5$).

Proposition 7.4 (Pairwise caps at $\sqrt{3}$ and 2)**.** *Let $r \ge \frac{1}{2}$ and suppose $\varphi_1(r)/\pi$ is irrational. Then $\Gamma(F)$ is bipartite for every finite family F of r-slots, so $\alpha(\Gamma(F)) \ge |F|/2$ and every pairwise cap at distance r is at least $\frac{1}{2}$. This applies to $r = \sqrt{3}$ and $r = 2$, where $\cos\varphi_1 = \frac{5}{6}$ and $\frac{7}{8}$ respectively.*

Proof. Assign to each slot $\{x, a\}$ the direction $\theta(\{x, a\}) \in \mathbb{R}/\pi\mathbb{Z}$ of the line through x and a. If $\{x, a\}$ and $\{x, b\}$ conflict, the triangle xab has sides $r, r, 1$, so the angle at x is φ_1 and $\theta(\{x, b\}) = \theta(\{x, a\}) \pm \varphi_1$ in $\mathbb{R}/\pi\mathbb{Z}$. Along a closed walk of length L in $\Gamma(F)$ we therefore get $(\varepsilon_1 + \cdots + \varepsilon_L)\varphi_1 \in \pi\mathbb{Z}$ with $\varepsilon_i = \pm 1$. Since φ_1/π is irrational, $\varepsilon_1 + \cdots + \varepsilon_L = 0$, so L is even. Hence $\Gamma(F)$ has no odd cycle and is bipartite, and the larger side of a bipartition is an independent set with at least $|F|/2$ slots. For $r = \sqrt{3}$ and $r = 2$, $\cos\varphi_1 = 1 - \frac{1}{2r^2}$ is rational and not in $\{0, \pm\frac{1}{2}, \pm 1\}$, so by Niven's theorem [16] φ_1/π is irrational. □

Thus at $\sqrt{3}$ and 2 a cap below $\frac{1}{2}$ needs constraints beyond pairwise slot conflicts. *Computational observation* (not certified): SAT computations on scaffolds of up to 45 points and exhaustive scans of chain-like configurations at $r = \sqrt{3}$ and $r = 2$ never gave a cap below $\frac{1}{2}$, while the same scans at $r = 1/\sqrt{3}$ found chains with cap 3/8.

7.3 The win condition

Before [17], the caps converted record-graph engineering into a quantitative attack on $\chi(\mathbb{R}^2) \geq 6$.

Theorem 7.5 (Win condition)**.** *Let G be a finite $\{1, \sqrt{3}\}$-graph with m unit edges such that every 5-coloring of its vertices that is proper on the $\sqrt{3}$-edges makes at least t unit edges monochromatic (any 6-chromatic $\{1, \sqrt{3}\}$-graph has $t \geq 1$). If $t/m > \frac{1}{3}$, then $\chi(\mathbb{R}^2) \geq 6$.*

Proof. Suppose c is a proper 5-coloring of $G(\mathbb{R}^2; \{1\})$. Scale G by $1/\sqrt{3}$: its $\sqrt{3}$-edges become unit pairs and its unit edges become pairs at distance $1/\sqrt{3}$. For every isometric placement of the scaled vertex set, c restricted to it is a 5-coloring proper on the former $\sqrt{3}$-edges, so by hypothesis at least t of the m former unit edges are monochromatic. Lemma 2.2 gives $p_{1/\sqrt{3}} \geq t/m > \frac{1}{3}$, contradicting Proposition 7.1. Hence no proper 5-coloring exists. □

The best certified value of the ratio is $t/m = 1/67$ (`certificates/forced-density/`; see the proof of Corollary 6.4): the 30-vertex graph of Theorem 5.2 with its 28 $\sqrt{3}$-edges and a minimal set of 67 of its 72 unit edges is already not 5-colorable, improving the ratio 1/72 of the full graph. The golden-ratio variant uses Proposition 7.2: a $\{1, \tau\}$-graph forcing unit-edge monochromatic density above $\frac{2}{5}$ under τ-proper 5-colorings would also have proved $\chi(\mathbb{R}^2) \geq 6$. The gap between 1/67 and 1/3 is large, and the question is now moot.

8 An extended six-coloring continuum

Following Mundinger, Pokutta, Spiegel and Zimmer [15], let X_6 be the set of $d > 0$ for which the plane has a coloring with six colors in which five color classes contain no two points at distance 1 and the sixth contains no two points at distance d. They showed $X_6 \supseteq [0.354, 0.657]$ (to three decimals). Reaching $d = 1$ would prove $\chi(\mathbb{R}^2) \leq 6$.

Theorem 8.1. $X_6 \supseteq [0.354,\ 0.667504532081]$.

The new part is $[657/1000,\ 667504532081/10^{12}]$; the rest is [15]. It is covered by 59 periodic tilings with rational vertex coordinates, each with eight polygonal tiles per period (five tile types for the five unit-avoiding colors and three for the sixth color), taken from a 33-parameter family without mirror symmetry. The certificates are in `certificates/six-coloring-continuum/`. For each tiling, the exact rational-arithmetic verifier `rigorous_verify.py` checks that all tiles are simple polygons, that periodic interiors do not overlap, that the tile areas sum exactly to the area of the lattice cell, that each of the five unit-avoiding tile types has diameter < 1 with distinct periodic copies more than 1 apart, and that the components of the sixth color have diameter below, and mutual separation above, an explicit rational interval of distances d. The sixth color then avoids every d in that open interval (with any deterministic assignment of boundary points). The script `verify_continuum.py` checks that the 59 intervals overlap and cover the stated range, and `verify_all.py` regenerates all 59 certificates from the stored parameter vectors before running the coverage check.

Remark 8.2 (Numerical observations). The following are floating-point observations and are not part of the proof of Theorem 8.1. The numerical optimum of the 33-parameter family is $d = 0.66750453208133\ldots$; there 23 constraints are active with positive multipliers and the reduced Hessian is numerically negative definite, indicating a strict local maximum for this tiling topology. Period doubling, 2×2, 3×1, 1×3 and 3×2 supercells, and all period-two relabelings of the five unit colors returned the same value.

9 Discussion and open problems

Open problems.

Problem 9.1. Is there a 25-vertex 6-chromatic two-distance graph? The 26-vertex graph is vertex-critical and further local search did not reduce it; no nontrivial lower bound on the minimum order is known for two-distance graphs.

Problem 9.2 (Finite 7-chromatic witness)**.** Realize the incidence structure of Proposition 4.1 by an actual finite $\{1, d\}$-point set, or answer Question 4.2. A finite $\{1, d\}$-graph W containing two points at distance $s \geq \frac{1}{2}$ that receive the same color in every proper 6-coloring would yield a finite non-6-colorable graph on fewer than $2|W|$ vertices: take two copies of W sharing one of the two points, rotated so that the two partner points are at distance 1.

Problem 9.3. Close the gap $99 \leq e_5 \leq 2257$ and $322 \leq e_6$. The tiling method of Section 6.2 appears exhausted for $k = 5$ (the regular hexagonal 4-coloring is a stationary point), and we know no other nontrivial lower-bound technique for the number of edges of 5-chromatic unit-distance graphs.

A Artifacts and reproducibility

All certificates are in the folder `certificates/` next to this manuscript; its `README.md` lists the check for each result, and all checks passed on 14 October 2026. Trusting an UNSAT claim requires trusting only the DRAT checker `drat-trim` [26] (available from `https://github.com/marijnheule/drat-trim`) and the short scripts that build or read the CNF files; trusting the SAT (colorability) claims requires only checking explicit colorings; the exact-arithmetic chain certificates additionally rest on `sympy`'s multivariate polynomial reduction and minimal-polynomial routines; the tiling certificates use only Python's exact `Fraction` arithmetic. The searches and the computational observations of Sections 4, 6 and 7 are not part of the bundle. Files ending in `.xz` must be decompressed first (`xz -dk`).

File(s)	Size	Content
`seven-colors/` (Theorem 1.1)		
`v8cert.cnf/.drat`	2.7 KB / 262 KB	proper 6-coloring of the V8 with $\leq$ 1 monochromatic diagonal: UNSAT (Proposition 3.2); 84 vars, 267 clauses.
`capA.cnf/.drat`	7.2 KB / 21 KB	proper 6-coloring of chain A ($r = \sqrt{2}$) with $\geq$ 6 of 11 slots monochromatic: UNSAT (Proposition 3.5); 188 vars, 646 clauses.
`capB.cnf/.drat`	7.2 KB / 21 KB	same for chain B ($r = 1 + \sqrt{3}$).

File(s)	Size	Content
`chain12.json`, `chain12b.json`, `v8.json`	<1 KB each	floating-point coordinates and designated pairs of the three configurations.
`verify_exact.py`	∼250 lines	exact symbolic certification of chains A and B (Section 3.3): tower-degree certificates via primitive-element minimal polynomials, all 66 pairwise distances per chain resolved in K_A resp. K_B, edge and slot sets confirmed against the float files, caps re-proved by exhaustive subset enumeration (dependencies: `sympy`, `python-sat`).
`g26/` (Theorem 5.1; $p_d \geq 1/37$)		
`deneve117_min.json`	<1 KB	the 26 integer vectors (Table 1).
`record26.cnf/.drat`	4.9 KB / 884 KB	5-coloring of the 26-vertex graph: UNSAT.
`verify117best.py`	∼60 lines	exact edges, non-5-colorability (two solvers), 6-colorability, vertex-criticality.
`g30/` (Theorem 5.2; $p_{\sqrt{3}} \geq 1/28$)		
`record30.cnf`, `record30.drat.xz`	5.8 KB / 1.3 MB	5-coloring of the 30-vertex graph: UNSAT.
`verify30.py`	∼70 lines	exact edges, non-5-colorability, 6-colorability, vertex-criticality.
`forced-density/` ($p_{1/\sqrt{3}} \geq 1/67$)		
`record67_graph.json`	1 KB	the 30 integer vectors and the 67 designated unit pairs.
`record67.cnf/.drat`	8.8 KB / 776 KB	5-coloring proper on all $\sqrt{3}$-pairs and the 67 unit pairs: UNSAT.
`verify67.py`	∼120 lines	exact distances, non-5-colorability, regeneration of the CNF and DRAT files, `drat-trim` check, minimality of the 67 pairs.
`e5-2257/` (Theorem 6.1)		
`edge2257_graph.json`	26 KB	edge list of the 2257-edge subgraph of the 509-vertex graph.
`edge2257.cnf.xz/.drat.xz`	17 KB / 27 MB	4-coloring of the subgraph (one triangle's colors fixed): UNSAT (sizes compressed).
`parts509_coordinates.vtx`, `check_unit.py`	—	exact coordinates of the 509 vertices; unit-distance check.
`e6-lower-bound/` (Theorem 6.3)		
`c5_best.json`	156 KB	the 4230 exact rational convex pieces, their colors and the period lattice.
`certify.py`	∼350 lines	exact tiling check and ball-arithmetic enclosure of p (dependency: `python-flint`).
`bounds.py`	∼30 lines	the conversion $p \mapsto (e_6, v_6)$.
`pmono.c`, `dump.py`	∼100 lines	independent floating-point evaluation of p.
`six-coloring-continuum/` (Theorem 8.1)		
`certificate_*.json`, `asym_*.npy`	—	the 59 exact tiling certificates and their parameter vectors.
`rigorous_verify.py`, `verify_all.py`, `verify_continuum.py`	—	exact check of one tiling; regeneration of all 59 certificates; interval-coverage check.

Reproduction: `pip install python-sat sympy numpy`; then, in the respective folders, `python3 verify_exact.py`, `python3 verify117best.py deneve117_min.json`, `python3 verify30.py`, `python3 verify67.py record67_graph.json` (which looks for `drat-trim` on the `PATH`, in the environment variable `DRAT_TRIM`, or as a second argument, and writes its CNF and DRAT files to the current directory), and `python3 verify_continuum.py`, and `python3 certify.py c5_best.json`; for any UNSAT certificate, `drat-trim X.cnf X.drat` and confirm `s VERIFIED`. All searches ran on a single laptop (Apple M4 Pro); total compute for the record graphs was minutes, for the edge trim and the 5-coloring hours.

Acknowledgments

We thank Jaan Parts, whose 2019 sketch [20] contains the skeleton of Theorem 1.1, and De Neve, Vanden Kerchove, Colle, Tavernier, and Pickavet, whose published coordinate tables [10] seeded the searches of Section 5. The searches, certificates and verification code were developed with the assistance of Claude (Anthropic).

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